Course Code	Name of the Course				
216U03E512	Control Systems and Applications				
Teaching Scheme	TH	P	TUT		Total
	03	--	--		03
Credits Assigned	03	--	--		03
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Dynamics, differential equations and Laplace transforms

Course Objectives:

Objectives of this course are: (paragraph)

- To understand the fundamental concepts of control systems and mathematical modeling of the system.
- To explore the concept of time response and frequency response of the system.
- To learn basics of stability analysis of the system.
- (To do)case study for any feedback system through activity based learning.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Derive the mathematical model of different types of the systems.
CO2. Understand the basics of state variable models of system.
CO3. Explore analysis of the systems in time domain.
CO4. Explore analysis of the systems in frequency domain.
CO5. Design a feedback system.

Detailed Syllabus

Module No.	Unit No.	Details	Hrs	CO
1		**Introduction to Control System Analysis**	12	CO1
	1.1	**Introduction:** Open loop and closed loop systems, feedback and feed forward control structure, control systems models and examples of control systems		
	1.2	**Dynamic Response:** Standard test signals, transient and steady state behavior of first and second order systems, steady state errors in feedback control systems and their types		
	1.3	**Transfer Function Models of Various Systems:** Models of mechanical systems, models of electrical systems, block diagram reduction, signal flow graph and Mason's gain rule		
2		**State Variable Models of LTI Systems**	08	CO2
	2.1	**State Variable Models of Various Systems:** State variable models of mechanical and electrical systems		
	2.2	**State Transition Equation:** Concept of state transition matrix, properties of state transition matrix, solution of homogeneous and non-homogeneous systems, concept of eigenvalues and eigenvectors, controllability and observability		
3		**Stability Analysis in Time Domain**	10	CO3
	3.1	**Concepts of Stability:** Concept of absolute, relative and robust stability, Routh stability criterion, lag and lead compensator #self-study: PID controllers		
	3.2	**Root Locus Analysis:** Root-locus concepts, general rules for constructing root-locus, Root-locus analysis of control systems		
4		**Stability Analysis in Frequency Domain**	10	CO4
	4.1	**Introduction**: Frequency domain specifications, response peak, peak resonating frequency, relationship between time and frequency domain specification of system and stability margins.		
	4.2	**Bode Plot:** Magnitude and phase plot, method of plotting Bode plot, stability margins on the Bode plots		
	4.3	**Nyquist Criterion:** Polar plots, Nyquist stability criterions, Nyquist plot, gain and phase margins		
5		**Applications of Control Systems**	05	CO5
	5.1	**Concepts and Block Diagrams:** Servo, Microcontroller, Robotic, Fuzzy, IoT, Communication systems and AI systems		
	5.2	**Introduction to Programmable logic controller (PLC):** ladder programming, architecture		
		Total	**45**	

***There will be IA as mini-project of 20 marks**

 # Students should prepare self-learning topics on their own. Self – learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA and or laboratory experiments.

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	M. Gopal	*Control Systems: Principle and Design*	Tata McGraw Hill, India	Fourth Edition, eighth reprint 2015
2.	Benjamin C. Kuo	*Automatic Control Systems*	Prentice Hall of India	Seventh Edition, 1995
3.	M.N. Bandyopadhyay	*Control Engineering.*	Prentice Hall of India	Second Edition, 2004
4.	J.Nagrath and M.Gopal	*Control System Engineering*	New Age International Publishers, India	Firth Edition, 2007